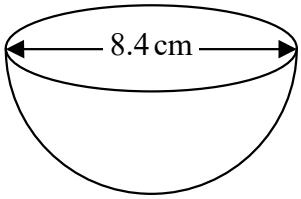
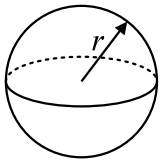
19 The diagram shows a hemisphere with diameter 8.4 cm.



Volume of sphere = $\frac{4}{3}\pi r^3$



Work out the volume of the hemisphere.
Give your answer correct to 3 significant figures.

Radius = 8.4 ÷ 2 = 4.2 cm.

Hemisphere volume = ½ × (4/3)πr³

= (2/3)π(4.2)³ ≈ 155.17 cm³.

To 3 significant figures: 155 cm³.

2 marks total
Use hemisphere formula
155 cm³ to 3 s.f.

..... cm³

(Total for Question 19 is 2 marks)

20 $d = \frac{1}{8}c^3$

$c = 10.9$ correct to 3 significant figures.

By considering bounds, work out the value of d to a suitable degree of accuracy.
Give a reason for your answer.

(Total for Question 20 is 4 marks)

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